# Synthesized Features

## A persona framework for collections allocation

Group Retail Risk · Group AI, Data & Analytics · November 2026

SUMMARY

# Synthesized features give allocation a view of the customer, built from data we already hold

## TODAY

No model beyond 60 DPD. Cases are allocated by bucket and product, and late buckets hold little behaviour to model.

## UC2 · SYNTHESIZED FEATURES

Signals from every data domain, condensed each month into four answers per account:

1. **How did they get here?**
2. **Can we reach them?**
3. **Will they pay?**
4. **Can they pay?**

## WHAT IT SHOWS

**54%** of card collections came from just a fifth of the accounts, picked out in advance by the framework.

**8×** Within the same DPD bucket (91–120), some loan customers pay 8 times as often as others.

**12%** of 180+ accounts show a positive signal; one in five of them pays, against under 3% of the rest.

## NEXT

**By 20 Nov**
UC2 synthesized features signed off; one may follow later

**20 Nov – 11 Dec**
UC1 transcript signals and UC2 combined, with action support

**11 – 20 Dec**
Pilot selection

**From 4 Jan**
Pilot

WHERE WE START

# Allocation follows the bucket, not the customer

WHO WORKS THE CASE

## In-house or agency

Rule-based: **DPD level and product** decide which team gets the case, not whether the customer can pay or has kept their word.

WHAT WE OFFER

## Settlement, plan or push back

Pre-approved settlement programmes (PLUS, CARES) with waivers fixed **by product and DPD stage**, not by the customer's capacity or track record.

WHERE 180+ EFFORT GOES

## 374k accounts at 180+

Worked as one pool, though a small share still shows signs they could pay. Finding them is the 180+ goal.

SYNTHESIZED FEATURES

# Everything we know about a customer, condensed into a few states

## DATA DOMAINS

- Card billing
- Card transactions
- CASA transactions
- Salary
- Loan instalments
- Bureau
- Digital logins
- Collector activity
- Customer records
- Transcripts · joining

→

## SIGNALS

- *Is salary regular?*
- *Do calls connect?*
- *Are promises kept?*
- *Last seen abroad?*
- *Paying while late?*

and 45+ more

→

## PERSONA STATES

**A1 · Deterioration**
How did they get here?

**A2 · Locatability**
Can we talk to them?

**A3 · Payment credibility**
Will they pay?

**A4 · Ability**
Can they pay?

UC1 conversations · joining

**10 data domains · 50+ signals · 4 states per account, every month**

61–180 DPD: personas that drive allocation

180+ DPD: find the customers with a positive signal

THE IDEA

# Three questions before a case reaches a collector

## 1 How did they get here?

A1 · Deterioration

- **Curing now** — stabilized
- **Cured or paid before** — slow roller · repeat delinquent
- **Paid, then stopped suddenly** — seasoned · early-life straight roller
- **Never paid** — early payment default

## 2 Can we talk to them?

A2 · Locatability

**Two checks**
where they were last seen · whether calls connect

- **Reachable** — calls connect
- **Non-Reachable** · calls fail
- **Skip** · last seen abroad

## 3 What can we offer?

The nuance, for every customer

**A3 · Credibility** · Will they pay?
High · Moderate · Low · Unknown

**A4 · Ability** · Can they pay? IN PROGRESS
High · Moderate · Low · Unknown

**UC1 · Conversations** · joining
intent and ability to pay, from call transcripts

INFORMS
Context and priority

INFORMS
How we can reach them

DECIDES
Offer and agent

HOW IT IS BUILT

# Built to be explained and trusted

### Rules you can trace

Every state comes from a written rule, and the record says which rule assigned it and why.

### Unknown is a state

When evidence is missing we say so, with the reason, instead of guessing.

### Point in time

Each month end uses only what was known that day: the view a strategy would have had.

### Customer-level evidence

Salary, CASA, bounces and logins are read across all of a customer's accounts, not only the delinquent one.

PART 1

# The four axes

Deterioration · Locatability · Payment credibility · Ability

A1 · DETERIORATION STATES · DECIDED BY RULES



# Customers curing now pay far more often than any other group

## HOW A CUSTOMER GETS A LABEL

Read from the DPD history alone, so it covers every account. Asked in order; the first "yes" decides.

| | | |
|---|---|---|
| 1 | Nothing paid since the first months? | → **Early payment default** bad from the start |
| 2 | Holding at one level for months? | → **Stabilized** stopped falling |
| 3 | Back within a year of curing? | → **Repeat delinquent** been here before |
| 4 | Fell to 61+ fast, after a year of good history? | → **Seasoned straight roller** sudden stop |
| 5 | Fell to 61+ fast, in the first year? | → **Early-life straight roller** fell fast, early |
| 6 | Otherwise | → **Slow roller** fell slowly, paying on the way |

## AUGUST MIX · SHARE OF ACCOUNTS




**60%** of card accounts curing now paid within 3 months; of those who only cured before, 22% did.

**Half** of loans (six in ten cards) paid for a while, then stopped all at once: an event, not a habit.

31 August 2026, 61–180 DPD · paid: meaningful payment within 3 months

A2 · LOCATABILITY STATES · DECIDED BY RULES



# Only one in five card customers at 61–180 can be reached

## HOW A CUSTOMER GETS A LABEL

### Location Where were they last seen?

Logins and in-person card use show where they are; a travel purchase or collector note signals they have left. The latest evidence wins.

In Country | Skip | Unknown

### Reachability Do our calls connect?

Recent call days all failed, no contact since → Non-Reachable. A recent contact → Reachable.

Reachable | Non-Reachable | Unknown

**Informs how we work a case:** by phone, or with tracing.

## AUGUST MIX

**Location × reachability, 61–180**

% of card accounts · loans below

| | Reachable | Non-Reachable | Unknown |
|---|---|---|---|
| In Country | 8.5% loans 10.2% | 13.6% loans 14.2% | 1.9% loans 2.7% |
| Skip | 6.2% loans 8.1% | 31.7% loans 28.5% | 6.1% loans 6.8% |
| Unknown | 6.3% loans 7.9% | 22.3% loans 14.9% | 3.4% loans 6.8% |

**1 in 3** look abroad and do not answer, the largest group on cards and loans alike: candidates for tracing or an agency.

**3.7%** of 180+ card accounts are reachable (loans 4.8%), a positive signal for finding those still worth working.

A3 · PAYMENT CREDIBILITY SCORE · POINTS → BAND



# Within the same bucket, credible customers roll far less

## HOW A CUSTOMER GETS A LABEL

Three sources earn points; we add them up

**Promises kept**
last 12 months
Kept +2 | Partly +1 | Broken −1

**Paying while delinquent**
regular = the monthly due paid in 2 of the last 3 months
Regular +2 | Some +1

**Record before**
clean = paid as billed before falling behind
Clean +1

| High | Moderate | Low |
|---|---|---|
| 4+ points | 2–3 points | 0–1 points |

> An explicit refusal to pay → **one band down**

## AUGUST MIX

| | High | Moderate | Low |
|---|---|---|---|
| Cards 61–180 | 2.5% | 19.4% | 78.0% |
| Loans 61–180 | 4.9% | 27.9% | 67.2% |
| Cards 180+ | 0.1% | 2.7% | 97.2% |
| Loans 180+ | 0.3% | 4.3% | 95.4% |

**42% vs 90%** of card customers at 91–120 DPD rolled further: High vs Low credibility.

**59%** of card customers made no promise to pay in the last 12 months: their credibility rests on payments alone.

Roll rates: back-test, 3-month outcomes compared within each DPD band

A4 · ABILITY IN PROGRESS



# Ability tells us what a customer can realistically be offered

## HOW A CUSTOMER GETS A LABEL

Four evidence families; we count how many are strong*

**Income**
Regular salary, ours or bureau

**Money in**
Money arriving or held, not salary

**Spending**
The customer's own spending

**Other products**
Activity on their other products

| High | Moderate | Low | Unknown |
|---|---|---|---|
| 2+ strong | 1 strong | some evidence, none strong | no evidence |

A cheque bounce or returned direct debit → **one band down**

## AUGUST MIX

| | High | Moderate | Low | Unknown |
|---|---|---|---|---|
| Cards 61–180 | 1.6% | 13.9% | 27.3% | 57.2% |
| Loans 61–180 | 1.2% | 8.8% | 49.3% | 40.7% |
| Cards 180+ | 0.3% | 6.0% | 8.6% | 85.1% |
| Loans 180+ | 0.1% | 3.5% | 11.0% | 85.4% |

**Needs: GRR's bureau salary feed**

About a third of customers have no CASA with us, and half never had a salary credit here. Without a regular bureau salary feed they stay Unknown; with it, income becomes visible across the book. Today the bureau's bulk salary pull reaches about 1% of customers.

**57%** of card customers at 61–180 show no evidence of their money yet.

**4×** High-ability card customers are back below 61 DPD three months later four times as often as those with no evidence.

* Strong = regular and above the floor in at least 2 of the last 3 months (floor: AED 3,000 or twice the amount due); at 180+: 3 of the last 6 months

PART 2

# From states to a strategy

Too many personas · only in-house needs a strategy · a first look

COMBINING THE STATES

# Combined, the states give hundreds of personas: far too many to give each a strategy

**6** deterioration states
A1

×

**9** locatability cells
A2

×

**3** credibility bands
A3

×

**4** ability bands
A4

=

**648** possible personas
today

×

**UC1** transcript signals
intent · hardship · dispute

**393**
occur among card accounts in August

**222**
have enough accounts (10+) to judge

A **persona** is one combination of states. Each UC1 signal multiplies the count again, and many personas hold only a handful of accounts.

Cards 61–180 · August 2026 · credibility High / Moderate / Low; ability includes Unknown

THE SIMPLIFICATION

# Only the cases we keep in-house need a strategy of their own

EVERY CASE

**61–180 DPD**

allocated first, then worked

→

**In-house** — STRATEGY PER PERSONA

Offer, agent and channel chosen for who the customer is

**Agency** — POLICY TERMS

Worked by the agency, paid on cash collected

**Non-workable pool** — POLICY TERMS

Least probable to pay, referred out under the policy

THE QUESTION CHANGES

FROM

~~A strategy for every persona~~

TO

**Which personas earn in-house effort, and what we do with them**

Agency and non-workable handling as in the collections policy (FR Policy on Operations, July 2025)

A FIRST LOOK

# How an account gets a tier: by what its persona actually paid

## 1 Describe each account

One state on each of the four axes. Together they form its **persona**.

- Stopped suddenly
- Skip · Non-Reachable
- Low credibility
- Unknown ability

## 2 Measure what it paid

For each persona: every account in May 2026, everything it paid in June–August, averaged per account.

**Example:** 400 accounts paid AED 70,000 in total → **AED 175 per account**. 10% of them paid, AED 1,750 each on average.

Observed, not modelled: the average combines how often and how much.

## 3 Rank and cut

The personas with at least 10 accounts (97% of cards), ranked by AED per account:

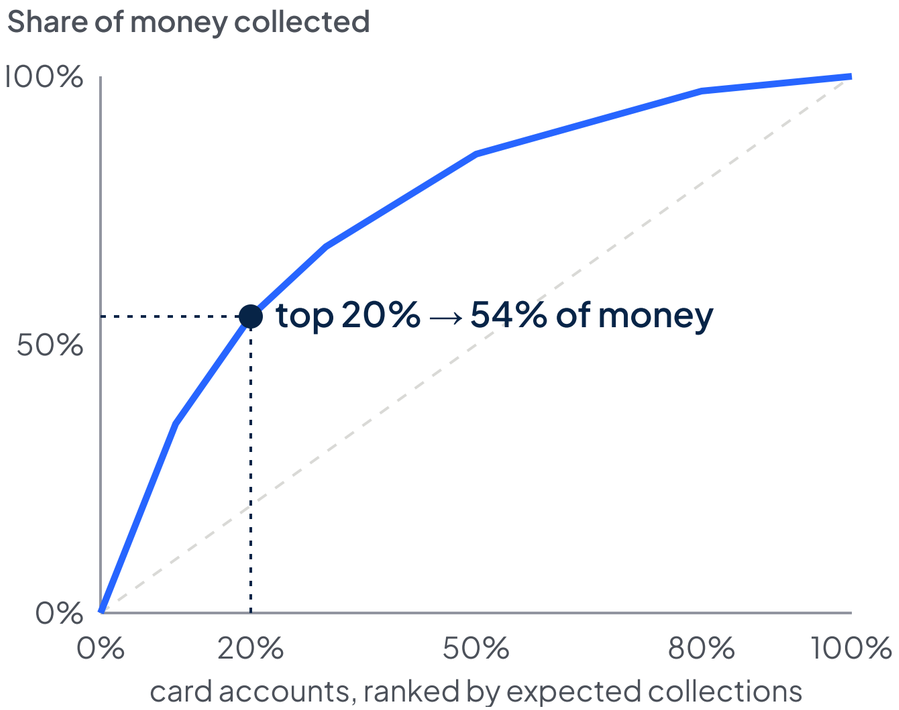
| | |
|---|---|
| In-house | AED 1,250 and above |
| Agency | between |
| Non-workable | under AED 175 |

Illustrative cut-offs from the data; GRR would set them from the commission rate card and in-house cost.

Cards 61–180 · back-test May → Jun–Aug 2026; tiers hold on unseen months · personas under 10 accounts unassigned (3%)

A FIRST LOOK

# In-house would take a sixth of card accounts and half the money collected




| Tier | Accounts | Money collected | Recovery, 3 m | Call days |
|---|---|---|---|---|
| **In-house** | 17% | 48% | 10.0% | 11.7 |
| **Agency** | 54% | 46% | 2.7% | 11.8 |
| **Non-workable** | 29% | 6% | 0.56% | 10.9 |

**In-house:** mostly customers curing now or cured before, who are credible or able.

**Non-workable:** mostly stopped suddenly or never paid, and not reachable.

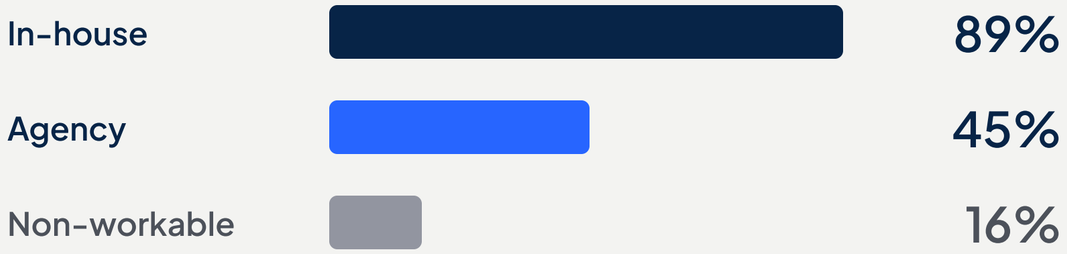
**10.9** call days per account in 3 months still go to the non-workable pool, almost as many as in-house.

Cards 61–180 · table out-of-sample (5 folds, Jan–May 2026); call days: August · cuts AED 1,250 / 175 · loans: same order, thresholds less stable

UC1

# In-house is where we talk to customers, so that is where UC1 adds its signals

RIGHT-PARTY CONTACT IN THE LAST 3 MONTHS · CARDS




All card accounts 41% · loans 52%

WHAT TRANSCRIPTS ADD

- Stated intent to pay
- Change in income or job
- Hardship reason
- Dispute
- Promise made on the call

> These come from what customers say on calls, so most of them will come from the in-house cases.

Cards 61–180 · August 2026 · groups at the illustrative cut-offs

TOWARDS THE STRATEGY WORKSHOP

# The in-house personas become a short list for GRR to give each a strategy

THE FIVE LARGEST IN-HOUSE PERSONAS · CARDS

| PATH | WHERE · REACH | CREDIBILITY | ABILITY | SHARE |
|---|---|---|---|---|
| **Stopped suddenly** | In country · **reachable** | Low | Unknown | **8.0%** |
| **Cured before** | In country · **reachable** | Low | Low | **6.4%** |
| **Curing now** | Unknown · **reachable** | Moderate | Unknown | **5.0%** |
| **Stopped suddenly** | In country · **reachable** | Moderate | Low | **4.9%** |
| **Cured before** | In country · **reachable** | Moderate | Low | **4.7%** |

**26** of 51 in-house personas cover 80% of in-house accounts

1 **The data proposes**

In-house personas, ranked by size and what they pay. All reachable; credibility and ability vary

2 **UC1 refines**

Transcript signals on top, where we talk to customers

3 **GRR decides**

Offer, agent and channel for each persona on the list, in the strategy workshop

Cards 61–180 · August 2026 · share of in-house accounts · in-house at the illustrative cut-off of AED 1,250 per account in 3 months · loans: 11 of 16 cover 80%

APPENDIX

# Signals, rules and governance

For the detailed discussion: where each signal comes from, the exact rules, and how the framework is controlled.

APPENDIX · A FIRST LOOK

# What each group could mean, for discussion

**In-house**

Curing now or cured before, reachable, credible or able

## Convert through skilled collectors: settlement, restructure or deferral

Policy tools: CARES settlements, loan restructure, deferral

**11% of cards · AED 40m**

50% paid in 3 months

**Agency**

Stopped suddenly or cured before, not reachable

## Place with agencies at 61–180, skip tracing where they look abroad

Policy: referral by product, DPD stage, contactability and strategy

**54% of cards · AED 213m**

17% paid in 3 months

**Non-workable**

Stopped suddenly or never paid, not reachable

## Free collector time: refer to agencies or stop calling

Policy: non-workable pool to agencies (HOD approval), or no calling with dunning as for Dead Beat accounts

**32% of cards · AED 138m**

4% paid in 3 months

180+ The 12% with a positive signal: lift out of the pool for an in-house attempt. About one in five pays, against under 3% of the rest.

For discussion in the strategy workshop; not agreed with GRR · cards 61–180; AED = outstanding principal; paid = meaningful payment

APPENDIX · A FIRST LOOK

# How a pilot could test it, if the strategy workshop takes it forward

## Test in each tier

- **In-house:** skilled collectors and settlements vs business as usual
- **Agency:** placement at 61–180 vs in-house calling
- **Non-workable:** agency referral or dunning only vs business as usual (hold-out)
- Random split within each tier, 3 months

## Measure

- Meaningful payment and roll rate
- Amount collected per account
- Recovery of outstanding principal
- Call days and cost per AED collected
- 180+: the flagged group vs the rest

## Size and guardrails

- Cards: agency and non-workable fill both arms in one month
- In-house needs a 10-pt target; loans need 10 pts and pooling
- Same policy and compliance rules
- Monthly readout and a stop rule

Sample per arm (cards, 5-pt lift): agency 970, non-workable 375; in-house 388 at a 10-pt lift · two-sided α 0.05, 80% power

APPENDIX · RISK SCORE

# Personas rank payers far better than the bucket; a model adds a little on top

How well each view ranks who pays (AUC) · 61–180

| Ranking customers by | Cards | Loans |
|---|---|---|
| DPD bucket only (today) | 0.65 | 0.62 |
| **Personas** | **0.83** | **0.77** |
| Personas × DPD bucket | 0.84 | 0.77 |
| Simple model (axis states + bucket) | 0.84 | 0.78 |

Out-of-sample: 5 folds by account, January–May 2026 cohorts. 0.5 = chance, 1.0 = perfect.

**Personas beat the bucket in every test**

+0.17 on cards, +0.14 on loans. The top 20% of card accounts by persona hold 54% of the money collected (bucket: 30%).

**The bucket adds almost nothing within personas**

+0.01 on cards, +0.00 on loans.

A model on the same inputs adds about **+0.02** over the personas. Risk score modelling is planned for **Phase 2**.

Paid = meaningful payment within 3 months (the risk score's target) · strict and rolling time splits give the same picture

APPENDIX A

# Signals and their sources

| Axis | Signals | Sources |
|---|---|---|
| A1 Deterioration | Monthly DPD bucket path, restructures linked to their original loans | Way4 billing history, loan master |
| A2 Locatability | UAE vs foreign logins, in-person card use (POS, ATM) by country, departure notes, travel purchases, call outcomes, deceased flags | Digital login, Way4 card transactions, CASA card transactions, DCORE follow-ups, customer records |
| A3 Credibility | Promises and plans with their outcome, payments vs dues per cycle, record before delinquency, refusals, disputes | DCORE plans and follow-ups, card statements and payments, loan monthly dues and cash |
| A4 Ability | Salary per month (larger of ours and bureau), CASA balance and activity, external card payments, cheque / direct-debit returns, bureau bounces | Wealth-management salary and CASA features, AECB salary and bounce data, loan master (instalment) |

Customer-level signals are read across every account of the customer (CIF). Windows end at the observation month end: 3 months at 61–180 DPD, 6 at 180+.

All sources are existing bank datasets; no new data feed is needed to run the framework

APPENDIX B

# The scoring rules at a glance

| Axis | Evidence and points | State |
|---|---|---|
| A1 Deterioration | DPD history only, from January 2024 (reliable window) | First “yes” of six ordered questions (Appendix C) |
| A2 Locatability | Latest location evidence: foreign → Skip; UAE within 3 months (6 at 180+) → In Country; travel rule. Last 3 call days failed (6 at 180+) → Non-Reachable | Location × reachability (9 cells) + Deceased override |
| A3 Credibility | Promises: Kept ≥ 80% +2, Partly +1, Broken < 40% or last 3 broken −1. Payments: monthly due paid in 2 of the last 3 months +2, any payment +1. Record before: 80% paid as billed +1 | High ≥ 4 · Moderate ≥ 2 · else Low; refusal −1 rung; no delinquency history → Unknown |
| A4 Ability | Four evidence families: income (salary, ours or bureau), money arriving or held, own spending, other products. In progress | High: 2+ strong · Moderate: 1 strong · Low: evidence, none strong; distress −1 band; no evidence → Unknown |

Full specification and reference workbook per axis

APPENDIX C

# How a journey state is assigned

| # | Question | Yes when | State |
|---|---|---|---|
| 0 | Start of journey seen? | Oldest month already delinquent and not stabilized | Unknown |
| 1 | Early payment default? | ≤ 3 months of history, rolled in, never paid since | EPD |
| 2 | Stabilized? | Pays about one month's due in 5 of the last 6 months (61–180 only) | Stabilized |
| 3 | Repeat delinquent? | Cured an earlier spell that reached 31–60 DPD, back within 12 months | Repeat delinquent |
| 4 | Straight roller? | Current to 61+ DPD in ≤ 3 months, no payment on the way | Seasoned (12+ months on book) or early-life |
| 5 | Otherwise | Paid something on the way down, kept rolling | Slow roller |

Billing artefacts repaired first and counted · retail loans on the same tree, buckets shifted by one

APPENDIX · ABILITY

# New ability rule vs the old income + CASA rule

SHARE OF ACCOUNTS IN EACH STATE, AUGUST

| | High | Moderate | Low | Unknown |
|---|---|---|---|---|
| Cards 61–180 | 1.62%<br>was 0.23% | 13.9%<br>was 14.4% | 27.3%<br>was 28.9% | 57.2%<br>was 56.6% |
| Loans 61–180 | 1.18%<br>was 0.28% | 8.8%<br>was 8.9% | 49.3%<br>was 44.7% | 40.7%<br>was 46.1% |
| Cards 180+ | 0.27%<br>was 0.03% | 6.0%<br>was 6.2% | 8.6%<br>was 6.4% | 85.1%<br>was 87.4% |
| Loans 180+ | 0.12%<br>was 0.03% | 3.5%<br>was 3.5% | 11.0%<br>was 6.9% | 85.4%<br>was 89.5% |

NEXT 3 MONTHS, 61–180*

| | Cards | | | Loans | | |
|---|---|---|---|---|---|---|
| State | Cure | Pay | Roll | Cure | Pay | Roll |
| High | 31% | 59% | 53% | 28% | 75% | 59% |
| Moderate | 16% | 41% | 73% | 15% | 65% | 69% |
| Low | 11% | 29% | 82% | 10% | 45% | 82% |
| Unknown | 8% | 20% | 88% | 9% | 36% | 85% |

**High is reachable:** 2 of 4 strong families, not a CASA balance the bank sweeps.

**Other products with us** give evidence for customers with no CASA activity.

**Only proven transactions count;** bounces and reversals cancel out.

* Cure: below 61 DPD at month 3 · Pay: any payment, even partial · Roll: moves to a worse bucket

APPENDIX D

# Status and sign-off

| Axis | Status |
|---|---|
| A1 Deterioration | **Closed** · signed off by GRR |
| A2 Locatability | **Closed** · signed off by GRR |
| A3 Credibility | Sign-off by 20 Nov |
| A4 Ability | Sign-off by 20 Nov, or later |
| UC1 transcripts | Joining, 20 Nov – 11 Dec |

Status as of 11 Nov 2026